ENR 1.6 RADAR SERVICES AND PROCEDURES

1 Primary Radar

1.1 Supplementary services

1.1.1 A radar unit normally operates as an integral part of the parent ATS unit and provides radar service to aircraft to the maximum extent practicable, to meet the operational requirement. Many factors, such as radar coverage, controller workload and equipment capabilities, may affect these services and the radar controller shall determine whether he is able to provide, or continue to provide, radar services in any specific case.

1.1.2 A pilot will know when radar services are being provided, because the radar controller will use the following phraseology:... (aircraft callsign) this is... Area / Approach, you are radar identified / or radar contact

a) Passing... (beacon);
b) ... (radial / DME) from ...;
c) executing radar turns VOR radial / DME ... from ... (station).

1.1.3 Radar facilities are established at the following ATC Centres in the RSA:-
Cape Town / Cape Town International Airport
Durban / King Shaka International Airport
East London / King Phalo Airport
Johannesburg / O R Tambo International Airport
Port Elizabeth / Chief Dawid Stuurman International Airport
Bloemfontein/Bram Fischer International Airport
George/George Airport

1.2 The application of radar control service

Radar coverage in controlled airspace at each of the centres mentioned in para 1.1.3 above is defined as per the Radar Minimum Altitude Charts in Aerodrome section of the AIP.

1.2.1 Controlled airspace

Radar identification is achieved according to the provisions specified by ICAO.
Radar control service is provided in controlled airspace to aircraft operating within radar range. This service may include;

a) Radar separation of arriving, departing and enroute traffic;
b) radar monitoring of arriving departing and enroute traffic to provide information on any significant deviation from normal flight path;
c) radar vectoring when required;
d) assistance to aircraft in emergency;
e) assistance to aircraft crossing controlled airspace;

f) warnings and position information on other aircraft considered to constitute a hazard;

g) information on observed weather conditions. when vectored by the radar controller, pilots will be assigned levels which will provide terrain clearance.

1.2.2 Uncontrolled airspace

When an aircraft which has been provided with radar control service leaves controlled airspace, the control service will automatically be terminated without the pilot being notified. However, ATC may provide either a radar advisory service or a radar information service in advisory or information airspace respectively. These services will only be provided at the discretion of the controller, in which case the pilot will be notified of the type of service provided.

1.3 Radar and radio failure procedures

1.3.1 Radar failure

In the event of a radar failure or loss of radar identification, instruction will be issued to restore non-radar standard separation and the pilot will be instructed to communicate with the parent ATS unit.

1.3.2 Radio failure

The radar controller will establish whether the aircraft radio receiver is working by instructing the pilot to carry out a turn or turns. If the turns are observed, the radar controller will continue to provide radar services to the aircraft.

If the aircrafts radio is completely unserviceable, the pilot should carry out the procedures for radio failure in accordance with local provisions. If radar identification has already been established, the radar controller will vector other aircraft clear of its track until such time as the aircraft leaves radar coverage.

2 Secondary surveillance radar (SSR)

For the present only, Bloemfontein/Bram Fischer International Airport, George/George Airport, Johannesburg / O R Tambo International Airport, Cape Town / Cape Town International Airport, Durban/King Shaka International Airport, Port Elizabeth/Chief Dawid Stuurman International Airport and East London/King Phalo Airport is equipped to provide secondary surveillance radar (SSR) service.

2.1 Emergency procedures

Except when encountering a state of emergency, pilots shall operate transponders and select modes and codes in accordance with ATC instruction. In particular, when entering Johannesburg FIR, pilots who have already received specific instruction from ATC concerning the setting of the transponder shall maintain the setting until otherwise instructed.

2.1.1 If the pilot of an aircraft encountering a state of emergency has previously been directed by ATC to operate the transponder on a specific code setting, this setting shall be maintained until otherwise advised. In all other circumstances the transponder shall be set to Mode A Code 77 (or 7700).

Note: A continuous monitoring of responses is maintained on -

Code 77 (7700) Aircraft emergency
Code 76 (7600) Radio failure
Code 75 (7500) Hijack or other act of violence.

2.1.2 Notwithstanding the procedures in para 2.1.1 above, a pilot may select the mode and code of his choice whenever the nature of the emergency is such that this appears to him to be the most suitable course of action.

OPERATION OF SSR TRANSPONDERS
Within South African airspace pilots must select the Transponder to Standby before affecting an SSR code change and returning the transponder to ON/ALT. This action is required to prevent possible loss of displayed aircraft position/label information and possible misidentification of aircraft in automated South African Air Traffic Control (ATC) systems due to temporarily selection (while making the changes) of a code already in use.

2.2 Radio communication failure and unlawful interference procedures

2.2.1 Radio communication failure procedure

In the event of a radio communication failure, a pilot shall select Mode A Code 76 (7600) and follow established procedures; subsequent control of aircraft will be based on these procedures.

2.2.2 Transponder failure procedures
Failure before intended departure

If the transponder fails prior to departure and cannot be repaired locally, pilots shall:

a) plan to proceed as directly as possible to the nearest suitable aerodrome where repairs can be effected.
b) inform ATC as soon as possible preferably before the submission of a flight plan. When granting clearance to such aircraft, ATC will take into account the existing and anticipated traffic situation and may require the modification of the time of departure, flight level or route of the intended flight;
c) insert in item 10 of the ICAO flight plan under SSR, the letter N for complete unserviceability of the transponder, or in the case of partial failure, the letter corresponding to the remaining transponder capability as specified in ICAO Doc 4444, Appendix 2.

Failure after departure.

If the transponder fails after departure or en route, ATS Units will endeavour to provide for continuation of the flight in accordance with the original flight plan. In certain traffic situations this may not be possible, particularly when the failure is detected shortly after take-off. The aircraft may then be required to return to the departure aerodrome

or to land at another aerodrome acceptable to the Operator and to ATC. After landing, pilots shall make every effort to have the transponder restored to normal operation. If the transponder cannot be repaired, provision as given in paragraph 2.2.2 shall apply.

2.2.3 Unlawful interference procedure

2.3 System of SSR Code assignment

2.3.1 Unless otherwise directed by air traffic control, the last assigned identity (Mode A) code in flight shall be retained. After landing pilots to ensure that Mode A 2000 is selected. If no identity code is assigned , Mode A 2000 shall be selected and retained.

2.3.2 While operating within South African Airspace all aircraft shall be equipped and operate a serviceable Mode A and C transponder to prevent occurrences of AIRPROX when;

a) Operating within or transiting any controlled or advisory airspace as listed in paragraph 2.3.3
b) Operating within 2000FT above or below such controlled or advisory airspace;
c) Operating below and within 5NM (vicinity) of defined lateral limits of controlled or advisory airspace boundaries;
d) Any other airspace where the operation of transponders have been prescribed by the Director of Civil Aviation
e) Recreational microlights, gliders, paragliders and hang gliders operating in the vicinity of any controlled airspace shall remain clear of all controlled airspace boundaries, while maintaining full radio communication and are exempt from clauses b and c of paragraph 2.3.2 until the entire paragraph 2.3.2 has been revised.

2.3.3 An example of radiotelephony phraseology to be used by ATC on transferring aircraft at a FIR boundary -
ZSMDC SQUAWK ALFA 2000 Contact Johannesburg Area on FREQ 120,300 MHz.

Scheduled and non-scheduled IFR traffic will be allocated different discreet codes by ATC to avoid confusion, on the radio.

Note 1:
Only Bram Fischer (FABL), Cape Town (FACT), George (FAGG), O R Tambo (FAOR), King Shaka (FALE), Chief Dawid Stuurman (FAPE) and King Phalo (FAEL) is presently equipped with SSR. Advice on further centres will be made available when they are suitably equipped.

Note 2:
All aircraft operating within the undermentioned airspace are to be equipped with Mode A and C transponder equipment.

Cape Town FIR

Cape Town Upper Control Area (UTA)
Cape Town Control Area (CTA)
Cape Town Terminal Control Area (TMA)
Cape Town Control Zone (CTR)

George Control Zone (CTR)
George Terminal Control Area (TMA)
Port Elizabeth Terminal Control Area (TMA)
Port Elizabeth Control Zone (CTR)
East London Terminal Control Area (TMA)
East London Control Zone (CTR)
All class 'C' ATS routes within the Cape Town FIR

Johannesburg FIR

Johannesburg Control Area. (CTA)
Johannesburg Terminal Control Area. (TMA)
Johannesburg Control Zone. (CTR) - O R Tambo.
Kruger Mpumalanga Control Zone (CTR)
Kruger Mpumalanga Terminal Control Area (TMA)
Lanseria Terminal Control Area (TMA) above 7 500 FT
Polokwane Control Zone (CTR)
Polokwane Terminal Control Area (TMA)
Bloemfontein Control Area (CTA)
Bloemfontein Control Zone (CTR) - Bram Fischer.
Bloemfontein Terminal Control Area (TMA)
Durban Control Area (CTA)
Durban Terminal Control Area (TMA)
Durban Control Zone (CTR) - King Shaka
Kimberley Terminal Control Area (TMA)
All class C ATS Routes within the Johannesburg FIR

2.3.4 An active surveillance control (MSSR) service is provided by Bloemfontein Approach Control (ATC) within the Bloemfontein TMA/CTR and Kimberley TMA (at or above 7000 FT ALT) where the following is applicable to all aircraft:

Aircraft must be equipped with a serviceable mode A & C transponder when operating below and within 5NM (vicinity) of the lateral limits of the Bloemfontein TMA or Kimberley TMA, unless operating within the Special Rules Area (SRA) for non-transponder operations below the Bloemfontein TMA as described within the SA AIP ENR 2.2 paragraph 1.1 or below the Kimberley TMA as described within the SA AIP ENR 2.2 paragraph 1.2

2.4 AFI PARTICIPATING AREAS

2.4.1 AFI has established four (04) Participating Areas (PAs) within which specific codes are allocated to enable assignment to aircraft. The four PAs consist of the following FIRs with associated States:

a) PA East (12) – Addis Ababa (Ethiopia & Djibouti), Juba (South Sudan), Mogadishu (Somalia), Asmara (Eritrea), Nairobi (Kenya), Entebbe (Uganda), Kigali (Rwanda), Bujumbura (Burundi), Dar-es-Salaam (Tanzania), Seychelles (Seychelles), Antananarivo (Madagascar, Comoros & La Reunion), and Mauritius (Mauritius).
b) PA South (9) – Lilongwe (Malawi), Lusaka (Zambia), Luanda (Angola), Beira (Mozambique), Harare (Zimbabwe), Gaborone (Botswana), Windhoek (Namibia), Johannesburg (South Africa, Swaziland & Lesotho), and Cape Town (South Africa).
c) PA Central (3) – Kinshasa (D. R. Congo), Brazzaville (Congo Brazzaville, Cameroon, Central African Republic, Gabon, Equatorial Guinea, Sao Tome and Principe), and N'Djamena (Chad, Part of Cameroon & Part of Central African Republic).
d) PA West (6) – SAL (Cabo Verde), Dakar (Senegal, Mauritania, Gambia, Guinea Bissau, Mali & Côte D'Ivoire), Roberts (Guinea Conakry, Sierra Leone & Liberia), Accra (Ghana, Togo & Benin), Niamey (Niger, Burkina Faso & Part of Mali), and Kano (Nigeria)

2.5 CODE ALLOCATION

2.5.1 International codes shall be assigned in accordance with the following principles governing the originating regional code assignment method (ORCAM):

a) when an aircraft enters the AFI Region (either when departing or during flight), the first ATS unit concerned in that PA shall assign to it a four-digit code;
b) each flight will keep the original code assigned on entering the AFI region for the whole flight time within that PA;
c) normally a code change will be required at the time a flight crosses (leaving) the AFI region and/or PA boundary. However, in specific cases and by specific arrangements agreed between the ATS units affected during the continuation of the flight, the assigned code may be retained beyond the AFI region and/or PA boundary. (Please ensure that similar international codes are not shared within different PA's when making these arrangements!).

2.5.2 Domestic codes are allocated for use by flights that, throughout their flight, remain within the boundaries of the agreed area of use of such codes (normally within one FIR).

2.5.3 Domestic codes will be assigned to flights and used only within one FIR.

2.6 MONITORING OF THE PLAN

2.6.1 While full implementation of the CMP must inevitably be achieved gradually, it is expected that progressive development of ground facilities will allow in future an increasing number of ATS providers to adhere to the provisions foreseen in the plan.

2.7 CODE SERIES FOR SOUTH AFRICA

FIR	Domestic	International
FAJA	4000 - 4077	7300 - 7377
	2700 - 2777	
	0400 - 0477	
	3500 - 3577	

	Domestic	International
FACA	0500 - 0577	4700 - 4777
	5400 - 5477	
	3100 - 3177	
	4500 - 4577	

3 Differences

PANS RAC (DOC 4444 - RAC/501) Part X, paragraph 2.7.1, 2.8 and 5.2.
The following differences exist and have been registered with the International Civil Aviation Organization (ICAO).

3.1 For the information of those aircraft operating within the Johannesburg and Waterkloof TMAs unidentified unknown targets performing as expected for general aviation within the confines of the Johannesburg Special Rules Area will be deemed to be separated from aircraft operating within the aforementioned TMAs by procedures as opposed to radar.

3.2 Pilots operating below the TMAs in the Special Rules Area should therefore ensure that they do not exceed the applicable altitude restrictions as stipulated (Johannesburg Special Rules Area.)

3.3 By virtue of the fact that uncontrolled VFR aircraft operate in the various training areas, below the Cape Town TMA and transit to/from these areas from/to FACT/FAYP/ FALW/ FAWN/FASH and FAMY, radar separation and information provided to aircraft operating within the Cape Town TMA will only take known identified targets into account.

3.4 For information of those aircraft operating within the Cape Town TMA, unidentified targets performing as expected for general aviation within the confines of the various flying training areas will be deemed to be separated from aircraft operating within the aforementioned TMA by procedures as opposed to radar.

3.5 Pilots operating below the TMA, within the flying training areas and transiting to/ from, should therefore ensure that they do not exceed the applicable altitude restrictions as stipulated and also remain below the TMA whilst routing to/from the training areas.

3.6 An active surveillance control (MSSR) service is provided by George Approach Control (ATC) within the George TMA/CTR where the following is applicable to all aircraft:
Aircraft must be equipped with a serviceable mode A & C transponder when operating below and within 5NM (vicinity) of the lateral limits of the George TMA.

4 Graphic portrayal of area of radar coverage.

4.1 All radar clearance charts are available from the SACAA web-site and Aerodrome section of the AIP.

5 Promulgation of special rules applicable to radar separation and information

5.1 JOHANNESBURG FIR

By virtue of the fact that uncontrolled VFR aircraft, operation within the Virginia Flying Training Area and between the Virginia Flying Training Area and Virginia Airport, are to remain below the Durban TMA, radar separation and information provided to aircraft operating within the Durban TMA will only take known identified targets into account.

For information of those aircraft operating within the Durban TMA, unidentified targets performing as expected for general aviation within the confines of the Virginia Flying Training Area and Virginia Airport will be deemed to be separated from aircraft operating within the aforementioned TMA by procedures as opposed to radar.

Pilots operating below the TMA, within those areas mentioned above, should therefore ensure that they do not exceed the applicable altitude restriction as stipulated.

5.2 CAPE TOWN FIR

By virtue of the fact that uncontrolled VFR aircraft operate in the various training areas, below the Cape Town TMA and transit to/from these areas from/to FACT/FAYP/FALW/FAWN/FASH and FAMY, radar separation and information provided to aircraft operating within the Cape Town TMA will only take known identified targets into account.

For information of those aircraft operating within the Cape Town TMA, unidentified targets performing as expected for general aviation within the confines of the various flying training areas will be deemed to be separated from aircraft operating within the aforementioned TMA by procedures as opposed to radar.

Pilots operating below the TMA, within the flying training areas and transiting to/from, should therefore ensure that they do not exceed the applicable altitude restrictions as stipulated and also remain below the TMA whilst routing to/from the training areas.

6 UNIFORM IMPLEMENTATION OF 10 MIN LONGITUDINAL SEPARATION

6.1 Letters of Procedure have been signed with the following Air Traffic Service Providers, detailing the Longitudinal Separation applied.

Air Traffic Service Provider	Effective date	Remarks
FAJO / FMMM ACC -FIC	10/09/2001	Longitudinal separation10 minutes.
FAJO / FQMA Approach	02/12/1999	Longitudinal separation10 minutes.
FAJA ACC / FQMA APP	02/12/1999	All regional flights: 10minute
FAJA ACC East / FQMA APP	24/07/2001	All regional flights: 10minute
FAJA ACC / FQBR ACC	02/12/1999	All regional flights: 10minute
FAJO / FIMM APP / FIC	NIL	Reaction to proposed letter still awaited from FIMM. They have acknowledged receipt of same on the 24/01/2000.
FAJO / FNLU FIC	24/02/2000	Longitudinal separation: 20 minutes. At this stage no RNP development envisaged. Awaiting implementation of AORRA.
FAJO / FYWH ACC	24/02/2000	Longitudinal separation: 20 minutes. - Intercontinental flights.
FAJA ACC / FYWH ACC	24/02/2000	All regional flights: 10minutes.
FAJA ACC / FVHA ACC	02/12/1999	All flights: 10 minutes.
FAJA ACC / FBSK APP	02/12/1999	All flights: 10 minutes.
FAJA ACC / FDMS APP	02/12/1999	All flights: 10 minutes.
FAJA ACC / FXMM APP	02/12/1999	All flights: 10 minutes.
NOTE: Effective date reflects the current Letter of Procedure, in some instances there were two or three versions signed covering periods from the 02/12/1999 to24/02/2000 hence discrepancy in effective date.		

7 RADAR SEPARATION ASSURANCE

7.1 Separation assurance using ATS surveillance as a primary aid in the area control environment is implemented within the FAJA and FACA FIR above FL195 (19500FT). All transponder equipped aircraft operating in all routes within South Africa above FL195 (19500FT) are tracked by at least 2 different ATS surveillance sensors.

7.2 Consequently the following ATM elements need to be considered. ATS Surveillance Control Service will be automatically transferred from Approach or from adjacent Area Control Sectors, which will mean that the frequency change will not include a termination of ATS Surveillance control Service. Pilots are to take note that the En-route separation of 10 minutes at same level may be reduced to 10NM, demanding appropriate vigilance, R/T discipline and accuracy.